

LRE Thrust Chamber Modeling with EcosimPro

Master in Space Transportation Systems

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Academic Year 2024/2025



SAPIENZA
UNIVERSITÀ DI ROMA



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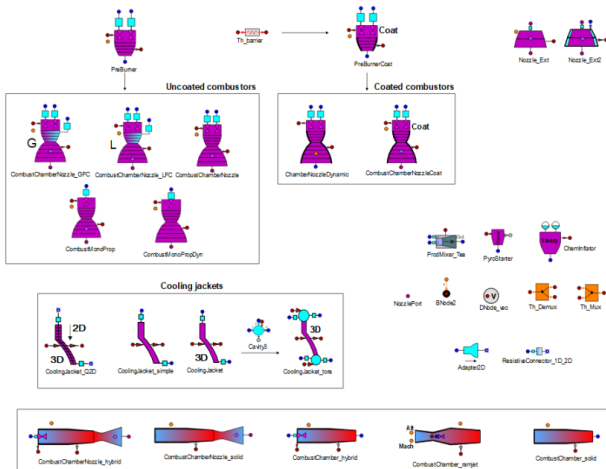
CJ Hands-on

► Steady State Modeling



Combustion Chamber library

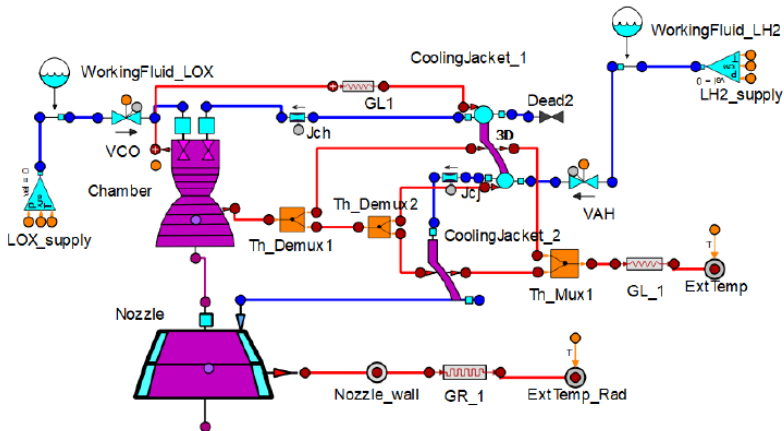
- The COMB_CHAMBERS library allows the modeling and simulation of liquid, solid and hybrid rockets





Combustion Chamber library

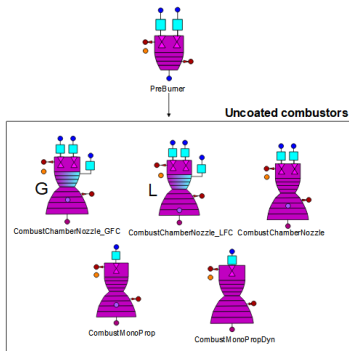
- Simulation of complex combustion chambers, comprehensive of regenerative cooling systems, is possible



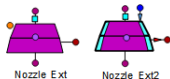


Combustion Chamber component

- The CombustChamber components allow the simulation of monopropellant and bipropellant liquid rocket combustion chambers



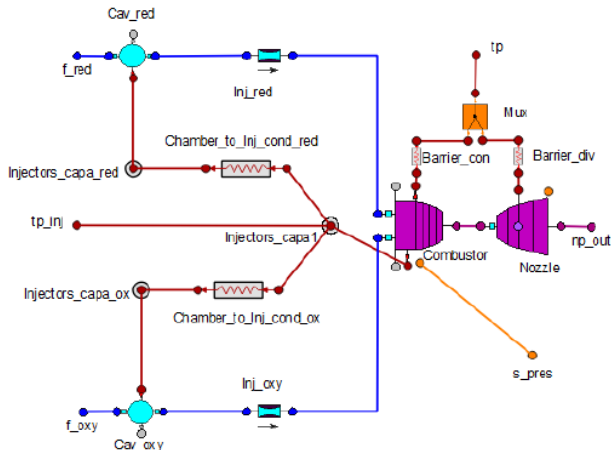
- Equilibrium/finite-rate chemistry
- Liquid droplets evaporation
- Liquid/gas film cooling models
- 1D non-isentropic nozzle model
- Heat transfer evaluated as Bartz + radiative heat flux
- Combustion products can be used by the FLUID_FLOW_1D library





Combustion Chamber

- All combustion chambers are build as "supercomponents", namely as the aggregation of several sub-components





Hands-on (1)

EcosimPro 2022 6.4.0 ENTERPRISE

File Edit View Tools Window Help

Simulation View

Workspace

CombustionChamber_tms.edi

Name	Version	Path
ELECTRICAL	3.2.0	C:/Users/win
EP	3.6.0	C:/Users/win
EVACPRO_UROM	1.1	C:/Users/win
EVACPRO_UROM_EX	1.0	C:/Users/win
FLUID_FLOW_1D	3.6.0	C:/Users/win
FLUID_FLOW_1D_EXAMPLES	3.6.0	C:/Users/win
FLUID_PROPERTIES	3.6.0	C:/Users/win
MASTER_STS	0.0	C:/Users/win
MATH	3.3.0	C:/Users/win
MECHANICAL	3.2.0	C:/Users/win
PLAYGROUND	0.0	C:/Users/win
PORTS_LIB	1.1.4	C:/Users/win
ROCKETS_EXAMPLES	3.6.0	C:/Users/win

Sym... Fil... It... Ext...

Name	Type
#LIBRARY_INFO	Library Info
_LIBRARY_MISC	Library Miscell
CC_DAY2	Component
CC_cooling_DAY2	Component
CombustionChamber_cooled_ss	Component
CombustionChamber_cooled_tms	Component
CombustionChamber_ss	Component
CombustionChamber_tms	Component
FULL_x2	Component
GG_sts15	Unknown
x2_tp_turbo	Component
x2_tps	Component
P_fed	Unknown
P_fed_cc	Unknown
Priming	Unknown
Priming_sts16	Unknown
Raptor_cc	Component
Raptor_cc_2	Component
Raptor_cc_complex	Component
V2_CombustionChamber	Component
Water_hammer	Unknown

Output

Messages Simulation Find Results



Hands-on (1)

MASTER_STS.CombustionChamber_trns - chamber

Type: COMB_CHAMBERS.CombustChamberNozzle

Name: chamber ☒ Show Label

Name	chamber	Units	Description
Nsub	5		Number of subsonic nodes; >2 if AUSM scheme !!
Dt	0.14	m	Nozzle throat diameter - also used to normalise diameters
Lc	0.5	m	Chamber length of subsonic part -also used to normalise 'Dc
Rcurv	0.01	m	Curvature radius at throat
Dc_vs_L	{{ 0,0.6,1},{ 2,2,1 }}	-	Subsonic diameters normalized with Dt vs. axial position nor
dx_c_input	FALSE		FALSE, dx_c_vs_L = weighting function for conv. mesh size dis
dx_c_vs_L	{{ 0,0.6,1},{ 1,1,1 }}	-	Weighting function for subsonic mesh size distribution or nc
emiss	0.01	-	Emissivity of the combustion gases; <0, calculated
Lrad	0	m	Mean beam length; 0, calculated as 0.9*D



Hands-on (1)

MASTER_STS.CombustionChamber_trns - chamber

Type: COMB_CHAMBERS.CombustChamberNozzle

Name: chamber ☒ Show Label

	Name	chamber	Units	Description
All Instances	Nsup	10		Number of supersonic nodes connected to the cooling jacket
Geometry	nozzle_type	NonIsentropic		Nozzle calculation option
Nozzle	Ld	1	m	Nozzle length to normalize 'Dd_vs_L' table
InitOptions	Ld2	1	m	Length from throat to nozzle exit, <= Ld
Scheme	Dd_vs_L	{{0,0,1,0,2,0,...,5,5,6,6,6,4}}	-	Supersonic diameters normalized with Dt vs. axial position normalized with Ld
ArrheniusData	dxd_input	FALSE		FALSE, dxd_vs_L = weighting function for div. mesh size distribution. TRUE, normalised node lengths vs node number
Injectors	dxd_vs_L	{{ 0,0,6,1},{ 1,1,1 }}	-	Weighting function for supersonic mesh size distribution or normalised node lengths vs node number
	AR_sup	0	-	Nozzle exit/throat area ratio: 0, no Dd_vs_L profile modification
	frozen_th	FALSE		Flag forcing frozen conditions in the throat
	frozen_nz	TRUE		Flag forcing frozen conditions in the supersonic nozzle
	Eta	0.95	-	Nozzle efficiency
	c_sf	0	-	Summerfield coefficient; = 0, no Summerfield correction
	GAM_ini	1.3	-	Initial gamma value, only to initialise nozzle conditions
	Floc[1,3]	{{ 0,0,0 }}	m	Location wrt the Mechanical Reference Frame centre O --i.e. wrt launcher interface at point O--
	Forient[1,3]	{{ 0,0,1 }}	-	Vector orientation not necessarily normalised

☒ Xref ☒ C-V = All Columns



Hands-on (1)

MASTER_STS.CombustionChamber_trns - chamber

Type: COMB_CHAMBERS.CombustChamberNozzle

Name: chamber

☒ Show Label

All Instances

Geometry

Nozzle

InitOptions

Scheme

ArrheniusData

Injectors

Name	chamber	Units	Description
P_o	-1	Pa	Initial chamber pressure - use a positive value to activate this data
T_o	-1	K	Initial chamber temperature - use a positive value to activate this data
P_ch	1000000	Pa	Deprecated data for the initial chamber pressure, not used if P_o > 0
T_ch	300	K	Deprecated data for the initial chamber temperature, not used if T_o > 0
Tcav_oxy	90	K	Initial oxidiser cavity temperature
Tcav_red	200	K	Initial reducer cavity temperature
MR_ini	0	-	Initial mixture ratio; > 0, ignited chamber initialisation
MR_max	20	-	Maximum mixture ratio allowing ignition - *10 for extinction
MR_min	0.05	-	Minimum mixture ratio allowing ignition - /10 for extinction
x_nco	0	-	Initial non-condensable mass fraction in cavities & combustor; =0 --> red vapors in combustor; <0 --> oxy vapors in co
Powder_comp	FLUID_PROP...wder_gases		Select Powder_gases, or a reduced SET_OF Chemicals belonging to BG_Option
starter_y[Powder_comp]	{ 0.26,0.19,0.07,0.21,0.27,0}	-	Mass fractions of the starter gases
eta	0.99	-	Combustor efficiency
rateOption	FALSE		TRUE, burning rate model active
tau_b	3e-05	s	Burning characteristic time - used when rateOption=TRUE
tau_c	0.003	s	Ignition time delay - used when rateOption=TRUE
GasLiqOption	UserDef		UserDef: user defined characteristic vaporization time
tau_v	0.0001	s	Vaporization characteristic time. Only if GasLiqOption=UserDef
D_dr_fu	1e-05	m	Fuel droplets size. Only if GasLiqOption=Advanced
D_dr_ox	1e-05	m	Oxidizer droplets size. Only if GasLiqOption=Advanced
hc_correl	Bartz		heat transfer correlation option
f_hc	1	-	Adjustment factor for the heat transfer correlation



Hands-on (1)

MASTER_STS.CombustionChamber_trns - chamber

Type: COMB_CHAMBERS.CombustChamberNozzle

Name: chamber ☒ Show Label

Name	chamber	Units	Description
scheme	centred		Numerical scheme used
K_u	0.25		AUSM velocity diffusion term coeff. for interface pressure
K_p	0.75		AUSM pressure diffusion term coeff. for interface Mach
lim	MinMod		Limiter used
order	first		Precision order in the reconstruction

6/25



Hands-on (1)

MASTER_STS.CombustionChamber_trns - chamber

Type: COMB_CHAMBERS.CombustChamberNozzle

Name: chamber

☒ Show Label

All Instances

Geometry

Nozzle

InitOptions

Scheme

ArrheniusData

Injectors

Name	chamber	Units	Description
BG_Option	FLUID_PROP....Chemicals		Select Chemicals in normal cases, or create a particular SET_OF of chemicals for reduced burned gases
reacts	1		considered reactions -- only used if Arrhenius = TRUE
Arrhenius	FALSE		
MR_min_Arrhenius	0.05		Minimum MR from which Arrhenius is applied
frz[Nsub]	1		=0, frozen flow at node i; =1, Arrhenius scheme at node i; =2, Delayed equilibrium at node i
mu_ij_r[BG_Option,reacts]	0		stoichiometric coefficient of species in reaction j among reactants
mu_ij_p[BG_Option,reacts]	0		stoichiometric coefficient of species in reaction j among products
mu_global_input	FALSE		TRUE, global reaction mechanism: coefficients for products are calculated; FALSE, no global reaction mechanism: coefficients for products are calculated from the equilibrium
mu_ij_r_global[BG_Option,reacts]	0		global coefficient of species in reaction j among reactants
Kfco[3,reacts]	0		Forward Rate values Arrhenius Formulation $K_f = A T^n \exp[-E_a/RT]$, see UM
Kbco_as_input	TRUE		TRUE, backward Rate values as input; FALSE, calculated from the equilibrium
Kbco[3,reacts]	0		Backward Rate values Arrhenius Formulation $K_b = A T^n \exp[-E_a/RT]$, see UM
calmo[reacts]	1		=1, Reaction number i active; =0, reaction number i non-active
rrv_on[reacts]	TRUE		Reverse reaction velocity flag
trdb_on[reacts]	FALSE		Third Body Activation flag
trdb[BG_Option,reacts]	0		Third body efficiencies



Hands-on (1)

MASTER_2023.COMBCHAMBER - chamber

Type: COMB_CHAMBERS.CombustChamberNozzle

Name: chamber

☒ Show Label

All Instances

Geometry

Nozzle

InitOptions

Scheme

ArrheniusData

Injectors

Name	chamber	Units	
Inj_redGases	FLUID_PROP...oBurnGases		Select noBurnGases in normal cases; Only for injectors do
Inj_oxyGases	FLUID_PROP...oBurnGases		Select noBurnGases in normal cases; Only for injectors do
mat_inj	None		Injector material
mass_inj	0	kg	= 0 calculated; Injector's mass
cond_inj	0.1	W/(m·K)	Injector thermal conductivity if material is None
capa_inj	500	J/(kg·K)	Injector specific heat capacity if material is None
th_inj_oxy	0	m	= 0 calculated; Characteristic wall thickness/width of oxi
th_inj_red	0	m	= 0 calculated; Characteristic wall thickness/width of redu
A_inj_oxy	0.00061	m ²	Oxydiser effective injection area
A_inj_red	0.00234	m ²	Reducer effective injection area
zetaf_inj_oxy	0.691	-	Oxi. injector forward pressure loss coefficient
zetaf_inj_red	2.73	-	Red. injector forward pressure loss coefficient
V_cav_oxy	1e-003	m ³	Oxydiser cavity volume
V_cav_red	1e-003	m ³	Reducer cavity volume



Hands-on (1)

MASTER_STS.CombustionChamber_trns - Jun_lox

Type: FLUID_FLOW_1D.Junction

Name: Jun_lox ☒ Show Label

All Instances

All

Name	Jun_lox	Units	Description
burnerGasesOption	FLUID_PROP...oBurnGases		Select Chemicals, or a reduced SET_OF Chemicals, only for compo
CV_option	FALSE		FALSE: pres. drop coef. formulation. TRUE: according to Cv formul
choked_option	TRUE		FALSE, Supersonic conditions allowed, no choked limitation
x_jun	0	m	Junction X coordinate relative to a body axis system
y_jun	0	m	Junction Y coordinate relative to a body axis system
z_jun	0	m	Junction elevation relative to a body axis system
Gcr_ideal	FALSE	-	FALSE: empirical critical flow correlations. TRUE (see UM §5.3.2.3): i
Ao	0.001	m^2	Throat area; if Ao=0, calculated as the minimal connected areas; =
lt	-1	m	Throat length. If equal to -1, assumed 3*D_throat
zetaf	0.1	-	Forward pressure loss coefficient; 0, calculated
zetaab	0.1	-	Backward pressure loss coefficient; 0, calculated
m_o	0	kg/s	Initial mass flow
Re_lam	2000	-	Laminar to turbulent transition Reynolds number
x_t	0.65	-	Pressure differential ratio factor without attached fittings at choked
Cv	0	gpm/psi^0.5	Valve flow coefficient; if Cv=0, calculated as 'sqrt(2)*A_min/2.4027
F p	1	-	Piping & internals flow area factor accounting for additional pressu



Hands-on (1)

MASTER_STG.CombustionChamber_trns - Jun_gh2

Type: FLUID_FLOW_1D.Junction

Name: Jun_gh2

☒ Show Label

All Instances

All

Name	Jun_gh2	Units	Description
burnerGasesOption	FLUID_PROP...oBurnGases		Select Chemicals, or a reduced SET_OF Chemicals, only for compone
CV_option	FALSE		FALSE: pres. drop coef. formulation. TRUE: according to Cv formulati
choked_option	TRUE		FALSE, Supersonic conditions allowed, no choked limitation
x_jun	0	m	Junction X coordinate relative to a body axis system
y_jun	0	m	Junction Y coordinate relative to a body axis system
z_jun	0	m	Junction elevation relative to a body axis system
Gcr_ideal	FALSE	-	FALSE: empirical critical flow correlations. TRUE (see UM §5.3.2.3): ide
Ao	0.005	m^2	Throat area; if Ao=0, calculated as the minimal connected areas; = 2.
lt	-1	m	Throat length. If equal to -1, assumed 3*D_throat
zetaf	0.1	-	Forward pressure loss coefficient; 0, calculated
zetaab	0.1	-	Backward pressure loss coefficient; 0, calculated
m_o	0	kg/s	Initial mass flow
Re_lam	2000	-	Laminar to turbulent transition Reynolds number
x_t	0.65	-	Pressure differential ratio factor without attached fittings at choked f
Cv	0	gpm/psi^0.5	Valve flow coefficient; if Cv=0, calculated as 'sqrt(2)*A_min/2.4027e-
F_p	1	-	Piping & internals flow area factor accounting for additional pressur



Hands-on (1)

BOUNDS

```
-- Set equations for boundaries: boundVar = f(TIME;...)
FLUID_FLOW_1D.Damp = 0.3
FLUID_FLOW_1D.GRAV = 9.806
FLUID_PROPERTIES.MinMolarFr = 1e-09
FLUID_PROPERTIES.VDW_option = 0
GH2_inlet.s_pres.signal[1] = 71E5
GH2_inlet.s_temp.signal[1] = 200
GH2_inlet.s_xNonCond.signal[1] = 0
LOX_inlet.s_pres.signal[1] = 71E5
LOX_inlet.s_temp.signal[1] = 90
LOX_inlet.s_xNonCond.signal[1] = 0
chamber.Combustor.IgnitFlag = 1
chamber.Combustor.f_v[1] = 1
chamber.Combustor.f_v[2] = 1
chamber.Combustor.f_v[3] = 1
chamber.Combustor.f_v[4] = 1
chamber.Combustor.f_v[5] = 1
chamber.Combustor.starter_T = 1000
chamber.Combustor.starter_m = 0
chamber.np_out.P = 30000
chamber.tp_inj.q[1] = 0
```

BODY



Hands-on (1)

- Run exp1
- The engine will reach 60 bar with a MR of 6.0
- From the monitor try
 - Bring the pressure to 45 bar retaining the same MR



Hands-on (1)

- Run exp1
- The engine will reach 60 bar with a MR of 6.0
- From the monitor try
 - Bring the pressure to 45 bar retaining the same MR
 - Lower MR to 5.5 keeping 45 bar of chamber pressure



Hands-on (1)

- Run exp1
- The engine will reach 60 bar with a MR of 6.0
- From the monitor try
 - Bring the pressure to 45 bar retaining the same MR
 - Lower MR to 5.5 keeping 45 bar of chamber pressure
 - Define a new custom variable Isp in seconds, not m/s



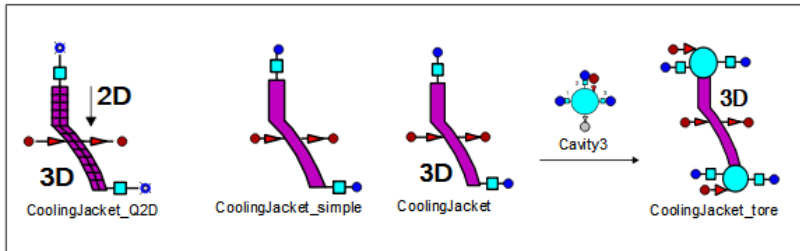
- Saving your results is fundamental!
- There are many ways:
 1. From the monitor, right click on a plot and export as .csv file
 2. In the experiment file, create a report table `REPORT_TABLE("results.csv", "*.m*")` -- create an automatic report of variables
Be sure to select only the variables you need or to use a large CINT interval, files can get pretty big!
 3. Save the entire state of the simulation (at `TIME=TSTOP` with
`SAVE_STATE("myState")`
You can use it as a starting point for your next run
`SET_INIT_ACTIVE(FALSE)`
`RESTORE_STATE("myState")`



Cooling Jacket Component

- The component represent a regenerative circuit of a chamber (with/without manifolds)
- Fluid model is either 1D or quasi 2D
- 3D wall geometry can be modeled

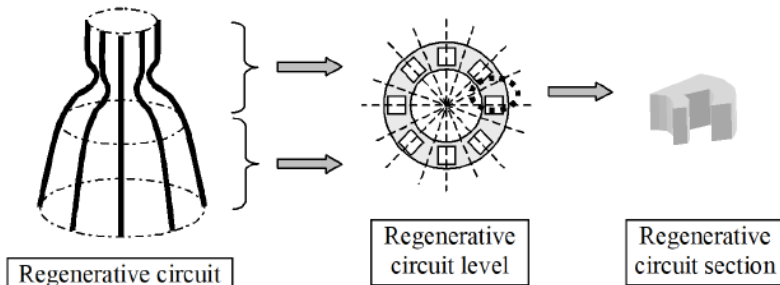
Cooling jackets





Cooling Jacket Component

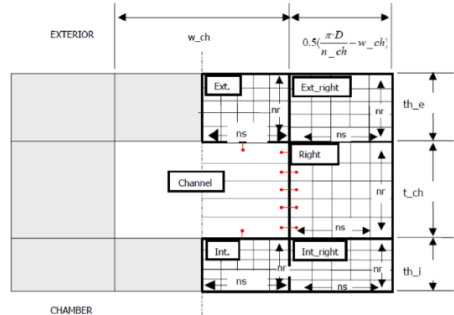
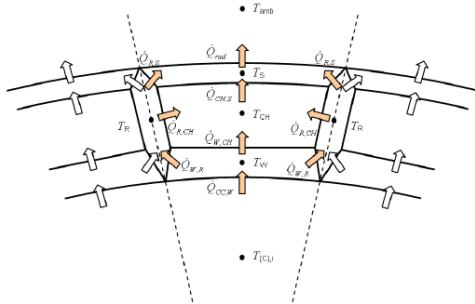
- The component represent a regenerative circuit of a chamber (with/without manifolds)
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Cooling Jacket Component

- The component represent a regenerative circuit of a chamber (with/without manifolds)
- Fluid model is either 1D or quasi 2D
- 3D wall geometry can be modeled





Hands-on (2)

EcosimPro 2022 6.4.0 ENTERPRISE

File Edit View Tools Window Help

Simulation View

Workspace

Name	Version	Path
COMP_DATABASE	3.4.0	C:/Users/win
CONTROL	5.0.3	C:/Users/win
ELECTRICAL	3.2.0	C:/Users/win
EP	3.6.0	C:/Users/win
EVACPRO_UROM	1.1	C:/Users/win
EVACPRO_UROM_EX	1.0	C:/Users/win
FLUID_FLOW_1D	3.6.0	C:/Users/win
FLUID_FLOW_1D_EXAMPLES	3.6.0	C:/Users/win
FLUID_PROPERTIES	3.6.0	C:/Users/win
MASTER_2023	0.0	C:/Users/win
MASTER_STS	0.0	C:/Users/win
MATH	3.3.0	C:/Users/win
MECHANICAL	3.2.0	C:/Users/win

Sym... Pl... It... Ext...

Name	Type
CC_DAY2.edi	Schematic
CC_DAY2.el	Source code
CC_I2.edi	Schematic
CC_cooling_DAY2.edi	Schematic
CombustionChamber_cooled_ss.edi	Schematic
CombustionChamber_cooled_trns.edi	Schematic
CombustionChamber_cooled_trns.el	Source code
CombustionChamber_ss.edi	Schematic
CombustionChamber_trns.edi	Schematic
CombustionChamber_trns.el	Schematic
FULL_I2.edi	Schematic
GG_STS_ESA.edi	Schematic
GG_sts15.edi	Schematic
I2_tp_turbo.edi	Schematic
I2_tps.edi	Schematic
P_fed.edi	Schematic
P_fed_cc.edi	Schematic
Priming.edi	Schematic
Priming_sts16.edi	Schematic
Raptor_cc.edi	Schematic
Raptor_cc_2.edi	Schematic

LIBRARY: MASTER_2023
FILE: cc_cooled
CREATION DATE: 07/04/2023

Output

Messages Simulation Find Results

Begin loading schematic:
C:/Users/winpropuser/EcosimPro_6.4.0/lib/MASTER_STS_library/MASTER_STS/schematics/combustion+chamber_cooled_trns.edi
End loading schematic

Begin saving schematic: C:/Users/winpropuser/EcosimPro_6.4.0/lib/ESPSS-v3.6/ws/MASTER_2023/schematics/cc_cooled.edi
End saving schematic



Hands-on (2)

MASTER_2023.cc_cooled - CoolingJacket

Type: COMB_CHAMBERS.CoolingJacket

Name: CoolingJacket ☒ Show Label

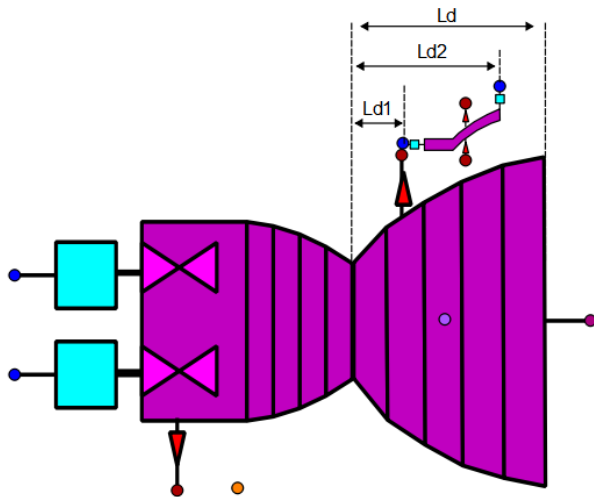
	Name	CoolingJacket	Units	Description
All Instances Geometry ChannelsWalls InitOptions	Nsub	5		Number of chamber fluid sections; =0 --> no convergent part is included
	Nsup	10		Number of nozzle fluid sections; =0 --> no divergent part is included
	Dt	0.14	m	Diameter to normalise 'Dc_vs_L' table
	Lc	0.5	m	Length of the convergent part -also used to normalise 'Dc_vs_L' tabl
	Ld	1	m	Nozzle length to normalize 'Dd_vs_L' table
	Ld1	0	m	Axial length from throat to nozzle inlet, 0 if Lc > 0
	Ld2	1	m	Axial length from throat to nozzle exit, <= Ld
	Dc_vs_L	{ { 0,0.6,1 }, { 2,2,1 } }	-	Subsonic diameters normalized with Dt vs. axial position normalized with Lc
	dx_c_input	FALSE		FALSE, dx_c_vs_L = weighting function for conv. mesh size distribution. TRUE, normalised node lengths vs node number
	dx_c_vs_L	{ { 0,0.6,1 }, { 1,1,1 } }	-	Weighting function for conv. mesh size distribution or normalised node lengths vs node number
	Dd_vs_L	{ { 0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1 }, { 1,1.1,1.5,2.1,2.7,3.5,4.3,5.5,6.6,6.4 } }	-	Supersonic diameters normalized with Dt vs. axial position normalized with Ld
	dx_d_input	FALSE		FALSE, dx_d_vs_L = weighting function for div. mesh size distribution. TRUE, normalised node lengths vs node number
	dx_d_vs_L	{{0,0.5,1},{1,1,1}}	-	Weighting function for div. mesh size distribution or normalised node lengths vs node number
	AR_sup	0		Nozzle exit/throat area ratio: 0, no Dd_vs_L profile modification

< XUL CgV All Columns >

Close



Hands-on (2)





Hands-on (2)

MASTER_2023.cc_cooled - CoolingJacket

Type: COMB_CHAMBERS.CoolingJacket

Name: CoolingJacket

☒ Show Label

All Instances

Geometry

ChannelsWalls

InitOptions

Name		CoolingJacket	Units	Description
nr	2			Number of radial thermal nodes at internal and external wall thickness
ns	2			Number of azimuthal thermal nodes between the channels
n_ch	120		-	Number of channels
w_ch	0.002		m	Channel width at throat section
t_ch	0.007		m	Channel height at throat section
rug	1.5e-006		m	Roughness
mat_e	Steel321			Chamber external material
th_e	0.005		m	Jacket outer wall thickness at throat section
K_e	0.1		W/(m-K)	External wall conductivity if mat=None
rho_e	1000		kg/m^3	External wall density if mat=None
cp_e	500		J/(kg-K)	External wall specific heat if mat=None
mat_i	Copper			Chamber internal material
th_i	0.0015		m	Jacket inner wall thickness at throat section
K_i	0.1		W/(m-K)	Internal wall conductivity if mat=None
rho_i	1000		kg/m^3	Internal wall density if mat=None
cp_i	500		J/(kg-K)	Internal wall specific heat if mat=None
wc_vs_L	{{ 0,0.5,1},{ 1,1,1 }}		-	Dimensionless channel widths vs dimensionless axial position
tc_vs_L	{{ 0,0.5,1},{ 1,1,1 }}		-	Dimensionless channel heights vs dimensionless axial position
te_vs_L	{{ 0,0.5,1},{ 1,1,1 }}		-	Dimensionless outer wall thickness vs dimensionless axial position
ti_vs_L	{{ 0,0.5,1},{ 1,1,1 }}		-	Dimensionless inner wall thickness vs dimensionless axial position



Hands-on (2)

MASTER_2023.cc_cooled - CoolingJacket

Type: COMB_CHAMBERS.CoolingJacket

Name: CoolingJacket ☒ Show Label

Name	Value	Units	Description
init_option	INIT_PT		Option to specify the initial thermodynamic state
P_o	0.1e5	Pa	Initial pressure
T_o	300	K	Initial temperature
rho_o	1	kg/m ³	Initial density
x_o	0	-	Initial quality
x_nco	0	-	Initial non-diluted, non-condensable gas mass fraction
ht_option	HT_critCond		Wall-cooling fluid heat transfer option
hc_dat	1000	W/(m ² ·K)	Heat transfer coefficient - only if ht_option = Ht_constant
k_f	1	-	Multiplier of the friction factor



Hands-on (2)

- Run exp1 with the same data of the previous simulation
- Which value of the GH2 inlet pressure is required to get the same pressure and MR?
- Try to use a valve sequence

```
TABLE_1D MFV_table = {{0, 2, 2.5, 100},{0,0,1,1}}
```

```
MFV.s_pos.signal[1] = timeTableInterp(TIME,MFV_table)
```



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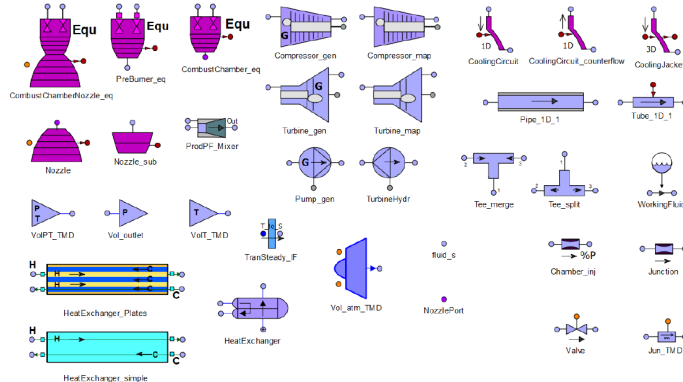
CJ Hands-on

► Steady State Modeling



STEADY LIBRARY

- The STEADY library allows to calculate the steady performances of LREs cycles under DESIGN and OFF-DESIGN conditions





STEADY LIBRARY

- The **DESIGN MODEL** contains the design conditions (some engines performances and imposed operating conditions) as part of the components input data. The experiment calculate the engine "size" and the rest of the nominal operating conditions.
- The **OFF-DESIGN MODEL** contain the initial conditions, actual sizing of the engine and the nominal operating conditions as part of the input data.



Type switches

- The ENUM type global variables define series of labels representing options or global constants to be used in the code:

NAME	DESCRIPTION	Values
DesignType	Design option	{ Design, Off_Design }
DesignType_Comb	Design option for Pre-burners	{ Design, Off_Design, Design_MR, Design_Tc }
PumpType	PumpType	{ known_Ns, known_flow, Off_D_pump }
TurbineType	TurbineType	{ known_mflow, known_PI_tt, known_pressures, Off_D_turb }
DeltaP	Valve DeltaP option	{ UserGiven, FromPorts, Percentage }
PumpStage	Pump number of stages & suctions	{ One, Two_S, Two_P }
RatioType	Mass flow ratio option in Tees	{ NoRatio, f2_vs_f1, f3_vs_f1 }
CombustorConfig	Constructional configuration of combustor	{ Rocket, AB_Annular, AB_Tubular, AB_CanAnnular }
TurbineConcept	Turbine assembly concept	{ turbopump, turbine }
PumpConcept	Pump assembly option	{ Electrical, TurbineDriven }
MASS_Estim	Mass calculation option for Vol_TMD components	{ knownPropMass, knownTBurn }
TankGeometry	Tank Geometry	{ Spherical, Cylindrical }
DesignHX	Heat Exchanger design option	{ full, dpPerc, eff }



Type switches

- Most components use some of these switches that enable switching the model between Design and Off-Design mode. The different options you can select are explained here:

DesignType = Design:

- Pipes, Junctions and Valves calculate their cross-sections from the calculated mass flow (from ports) and ΔP value either taken from the ports ("FromPorts" sub-option), or fixed from the user as a DATA ("UserGiven" sub-option). You need to determine which valves/Junctions are "UserGiven" or "FromPorts" (typically valves for turbine bypasses or adjusting a given pressure rise of a pump).
- Cooling circuits calculate the channel width and the internal wall thickness ("t_ch_cal" and "th_i_cal") at the location where the wall temperature is maximum (normally this occurs at throat position) of the Cooling Jacket from known wall temperature and ΔP .
- Combustion chambers calculate the inlet mass flows from given chamber pressure, mixture ratio (MR) and nozzle throat diameter (input data).

DesignType = Design_MR:

- Only for Pre-burners. Chamber pressure and mass flows are calculated from a given MR (input data) and from the calculated mass flows (from ports). This option is useful when the pre-burner pressure is determined by the downstream component, typically a turbine at given pressure ratio and known valves pressure drops.



Type switches

DesignType = Off_Design:

- Pipes, Junctions and Valves have a given geometry. Mass flow and ΔP are calculated by the model, depending on the problem.
- Cooling circuits have a given geometry and calculate ΔP and wall temperatures from the rest of the model variables, depending on the problem.
- Combustion chambers have a given nozzle throat diameter. Chamber pressure and MR are calculated from the rest of the model variables, depending on the problem.



Type switches

- PumpType:

PumpType = known_Ns (as full design conditions, no performance map)

- Pumps calculate needed torque and power from mass flow and pressures (taken from the ports), and efficiency (user given). Rotation speed is calculated from the specific speed N_s (input data).

PumpType = known_flow (as partial design conditions, no performance map)

- Pumps calculate needed torque and power from mass flow & pressures (taken from the ports) and efficiency (user given). Rotation speed is conditioned by another mechanically linked pump.

PumpType = Off_D_pum (as off-design conditions, use of performance map)

- Pumps calculate *current* rotation speed, torque and power from the pump nominal conditions (input data), map performances and from the rest of the model, depending on the problem.



Type switches

- TurbineType:

TurbineType = known_PI_tt (as full design conditions, no performance map)

- Turbines calculate mass flow (from ports variables), torque, rotation speed (from the mechanical port) and efficiency (user given). Pressure ratio is forced from the specified PI_tt value (input data).

TurbineType = known_mflow (as partial design conditions, no performance map)

- Turbines calculate mass flow (from ports variables), torque, rotation speed (from the mechanical port) and efficiency (user given). Pressure ratio is conditioned by the connected components (from port variables).

TurbineType = known_pressures (as partial design conditions, no performance map)

- The same as before but forcing the known inlet pressure when calculating the outlet turbine properties.

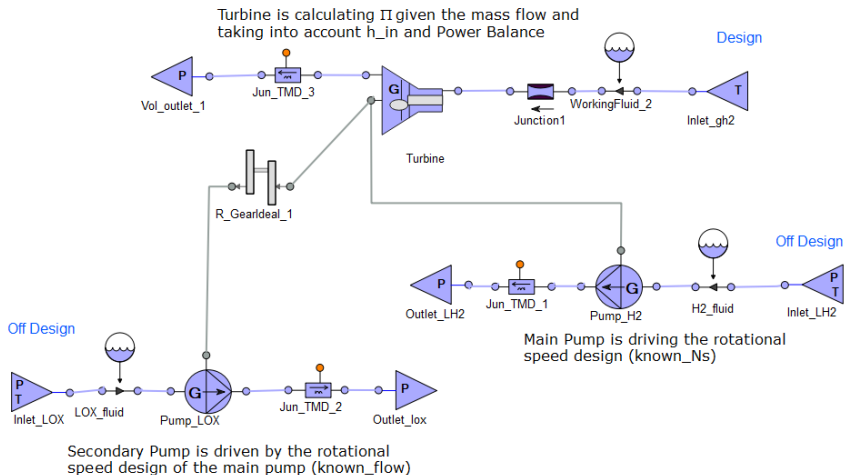
TurbineType = Off_D_turb (as off-design conditions, use of performance map)

- Turbines calculate *current* rotation speed, torque and power from the turbine nominal conditions (input data), performance map and from the rest of the model variables, depending on the problem.



Turbopumps - Hands On

- Design approach: in each line the mass flow is given to simulate infos coming from TC component

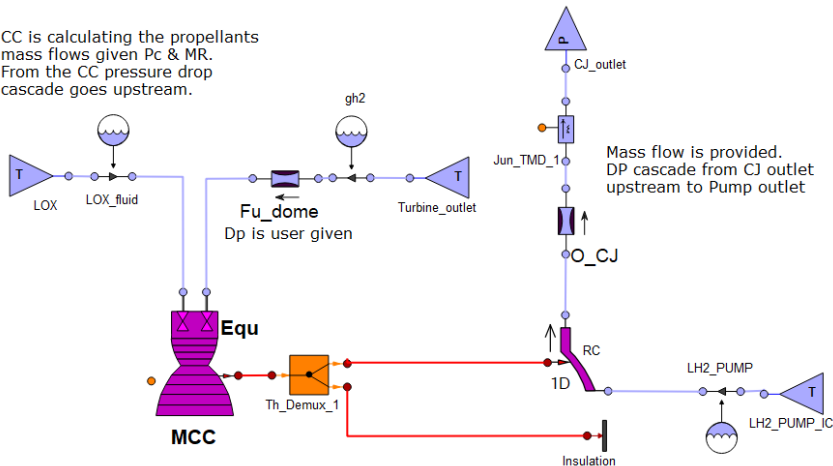




Thrust Chamber - Hands On

- Design approach: combustion chamber is the main component of the system. Mass flow infos propagate upstream to all components.

CC is calculating the propellants mass flows given P_c & MR .
From the CC pressure drop cascade goes upstream.





Useful equations

Pump

$$N_s = \frac{\omega Q^{1/2}}{\left(\frac{\Delta p}{\rho g}\right)^{3/4}}, \quad P_p = \eta_p \dot{m} \frac{\Delta p}{\rho}$$

Turbine

$$P_t = \eta_t \dot{m} c_p T_0 \left[1 - \left(\frac{p_e}{p_0} \right)^{(\gamma-1)/\gamma} \right]$$



Contacts

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This slides are based on previous work by

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- Marco Leonardi



LRE Thrust Chamber Modeling with EcosimPro

Thank you for listening!

Any questions?